

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, ETL Pipelines, Kafka, PySpark, scikit-learn, PyTorch, TensorFlow

• **Stack:** LangChain, LlamaIndex, Hugging Face, OpenAI API, FAISS, Pinecone, Prompt Engineering, Streamlit, MLOps

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable RAG and agentic AI system using LangChain, FastAPI, PostgreSQL, and vector embeddings with semantic search for production applications
- Designed efficient, fault-tolerant ETL pipelines with MLOps monitoring to process and index large document datasets for low-latency vector search and retrieval
- Improved system performance by **64%** through optimized indexing strategies and pipeline improvements in production environments
- Developed backend services enabling real-time, high-performance **AI model serving** of unstructured data across distributed systems

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
- Built production-ready Python generative AI pipeline with prompt engineering and Hugging Face transformer models, improving response accuracy by 43%
- Developed asynchronous AI pipelines integrating FAISS vector databases for real-time, fault-tolerant semantic retrieval workflows with focus on scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed scalable NLP and generative AI pipelines using OpenAI APIs and Python, reducing processing latency by 41% and improving output quality
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows to ensure **scalable integration** across distributed systems
- Implemented AI agent workflows and patterns using LangChain to automate large-scale data processing and enable autonomous task execution

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
- Automated recurring **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%** and improving consistency

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for feature prioritization

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built end-to-end ML system using Python, scikit-learn, and LightGBM to predict credit risk from time-series transaction data, covering feature engineering, model training, evaluation, and deployment
- Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency to improve model performance
- Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers